HW2 QUESTION2 REPORT

In this question, I used **Template Method Design Pattern** becasue all three different genetic algorithm versions have very similar operations. Differences occurs only in selection and cross over types. The other operations like **generateInitialPopulation(),**

computeFitness(),mutation() have same implementation for three different genetic algorithm versions.

So I make abstract these **selection**() and **crossOver**() methods in abstract ***GeneticAlgorithm*** class. Then I create **Version1Ga**, **Version2GA** and **Version3GA** concrete classes. These concrete classes extends from abstract ***GeneticAlgorithm*** class and implements the **selection()** and **crossOver** methods according to their versions.

I create a **Chromosome** class for storing datas and operations about a cromosome.

Also I use **Observer Design Pattern**. Becasue **DynamicPlot** objects are observers and **GeneticAlgorithm** is subject. Because **DynamicPlot** objects should be updated when new iteration is occured. So **DynamicPlots** objects register to **GeneticAlgorithm** and they can easily updated when new iteration is occured.

DynamicPlot class has a **GeneticAlgorithm** reference and **DynamicPlot** class uses the **GeneticAlgorithm** datas for drawing the plots.

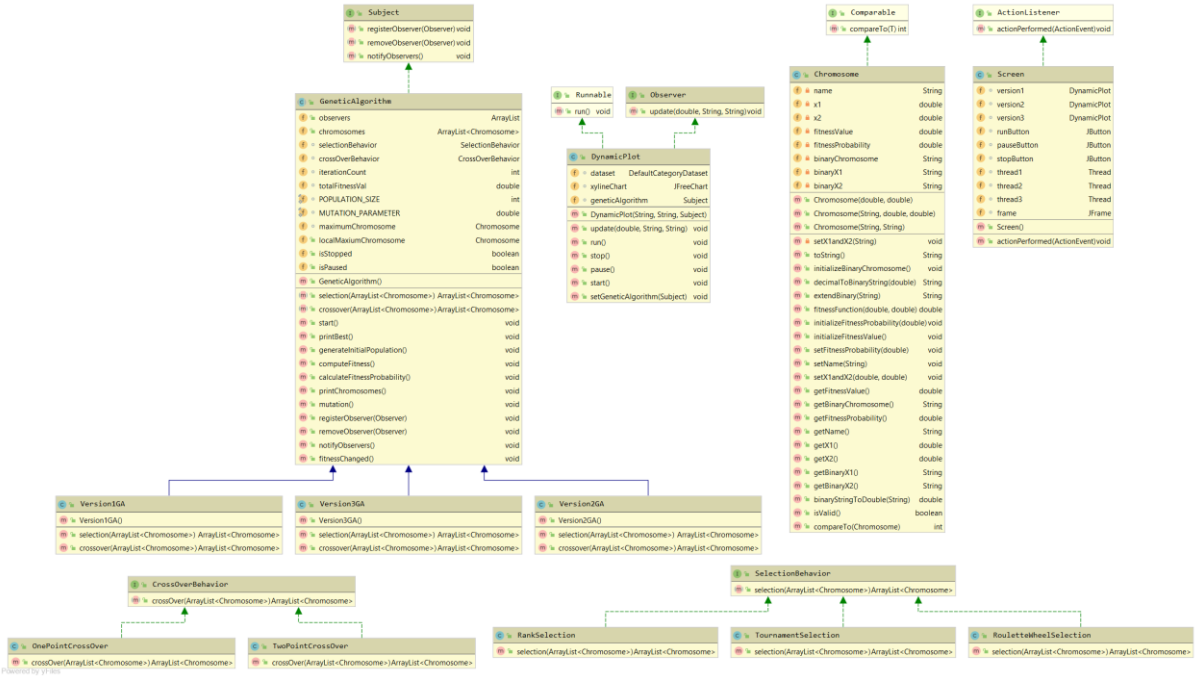
Screen class has three **DynamicPlot** for each of three versions of genetic algorithms.

I used three **threads** for drowing and updating the three **DynamicPlot** objects.

DynamicPlot object uses **GeneticAlgorithm** datas to update himself.

Also I use **Strategy Design Pattern** for Selection and Cross Over types to avoid the code duplication. I implements the Selection types by implementing the **SelectionBehavior** interface and I implements the Crossover types by implementing the **CrossOverBehavior** interface.

Class diagrams generated by Intellij IDE are in next page:



Burhan ELGÜN

151044060